

Supplementary Information

Radionuclide Identification for Plastic Scintillation Detectors by Backward Cumulative Channel Sum

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Supplementary Table S1. Measurement conditions for primary detector panel.

Data ID	Category	Radionuclide(s)	Nominal activity (μCi)	Distance (cm)	Actual live time (s)	Role / note
Background	Background-only	-	-	-	20.000	Background correction and resampling
1	Single-source	Am-241	90	80	20.000	Am-241 reference spectrum
2	Single-source	Am-241	90	180	20.000	Single-source distance measurement
3	Single-source	Am-241	90	280	20.000	Single-source distance measurement
4	Single-source	Am-241	90	460	20.000	Single-source distance measurement
5	Single-source	Cs-137	16	70	19.284	Cs-137 distance series
6	Single-source	Cs-137	16	130	19.652	Cs-137 reference spectrum
7	Single-source	Cs-137	16	300	19.900	Cs-137 distance series
8	Single-source	Cs-137	16	390	19.962	Cs-137 distance series
9	Single-source	Co-60	9	60	18.672	Co-60 distance series
10	Single-source	Co-60	9	120	19.220	Co-60 reference spectrum
11	Single-source	Co-60	9	310	19.720	Co-60 distance series
12	Single-source	Co-60	9	470	19.864	Co-60 distance series
13	Mixed-source	Cs-137 + Co-60	16 / 9	136 / 136	19.134	Mixed-source dataset
14	Mixed-source	Am-241 + Co-60	90 / 9	287 / 287	19.676	Mixed-source dataset
15	Mixed-source	Cs-137 + Co-60	16 / 9	361 / 361	19.716	Mixed-source dataset
16	Mixed-source	Am-241 + Cs-137	90 / 16	163 / 163	19.628	Mixed-source dataset
17	Mixed-source	Am-241 + Cs-137 + Co-60	90 / 16 / 9	309 / 309 / 309	19.574	Mixed-source dataset
18	Mixed-source	Co-60 + Am-241	9 / 90	153 / 210	19.310	Separated source positions
19	Mixed-source	Cs-137 + Co-60	16 / 9	367 / 285	19.568	Separated source positions
20	Mixed-source	Am-241 + Cs-137	90 / 16	85 / 173	19.554	Separated source positions
21	Mixed-source	Cs-137 + Co-60	16 / 9	223 / 316	19.470	Separated source positions

Notes. Activities are nominal source activities. Distances are measured from the detector surface with each sealed point source aligned to the horizontal and vertical center of the detector panel. Samples 13-17 used the same distance for all constituents; Samples 18-21 used separated source positions. The nominal acquisition time was 20 s, and all calculations used the actual live time recorded in each MCA file. For the primary detector panel, Channels 4–256 were treated as valid.

Supplementary Table S2a. Complete BCCS-NNLS results.

Sample	Expected	R _{net} (%)	Net cps	Am coefficient	Cs coefficient	Co coefficient	Residual (%)	Coefficient ranking
1	Am-241	23.53	1434.80	1.0000	0.0000	0.0000	0.00	Am-241 > (Co-60 = Cs-137)
2	Am-241	7.61	464.30	0.3157	0.0000	0.0054	5.78	Am-241 > Co-60 > Cs-137
3	Am-241	3.24	197.45	0.1288	0.0000	0.0000	26.57	Am-241 > Co-60 > Cs-137
4	Am-241	1.06	64.60	0.0423	0.0000	0.0000	78.98	Am-241 > Co-60 > Cs-137
5	Cs-137	132.28	8066.76	0.0000	2.5810	0.0000	1.35	Cs-137 > Co-60 > Am-241
6	Cs-137	52.00	3170.93	0.0000	1.0000	0.0000	0.00	Cs-137 > (Co-60 = Am-241)
7	Cs-137	12.62	769.69	0.0267	0.2197	0.0140	1.61	Cs-137 > Am-241 > Co-60
8	Cs-137	7.54	460.11	0.0551	0.1276	0.0000	6.61	Cs-137 > Am-241 > Co-60
9	Co-60	134.58	8206.81	0.0000	0.0000	2.7633	1.57	Co-60 > Cs-137 > Am-241
10	Co-60	49.50	3018.40	0.0000	0.0000	1.0000	0.00	Co-60 > (Cs-137 = Am-241)
11	Co-60	9.66	588.82	0.0282	0.0000	0.1827	1.35	Co-60 > Am-241 > Cs-137
12	Co-60	4.79	292.31	0.0237	0.0107	0.0745	3.84	Co-60 > Am-241 > Cs-137
13	Cs-137 + Co-60	89.72	5471.63	0.0323	0.9487	0.7973	0.48	Cs-137 > Co-60 > Am-241
14	Am-241 + Co-60	13.53	825.31	0.1405	0.0128	0.1954	1.31	Co-60 > Am-241 > Cs-137
15	Cs-137 + Co-60	16.85	1027.84	0.0606	0.1771	0.1284	1.20	Cs-137 > Co-60 > Am-241
16	Am-241 + Cs-137	44.56	2717.22	0.3575	0.6898	0.0050	0.53	Cs-137 > Am-241 > Co-60
17	Am-241 + Cs-137 + Co-60	24.68	1505.00	0.1718	0.2244	0.1879	1.31	Cs-137 > Co-60 > Am-241
18	Co-60 + Am-241	38.23	2331.37	0.2423	0.0000	0.6548	0.78	Co-60 > Am-241 > Cs-137
19	Cs-137 + Co-60	20.45	1247.37	0.0582	0.1636	0.2175	1.12	Co-60 > Cs-137 > Am-241
20	Am-241 + Cs-137	54.82	3343.30	0.9461	0.6093	0.0191	0.87	Am-241 > Cs-137 > Co-60
21	Cs-137 + Co-60	29.65	1807.91	0.0756	0.2129	0.3418	1.30	Co-60 > Cs-137 > Am-241

Notes. Reference rows are shaded. Coefficients are detector-response scaling factors relative to Samples 1, 6, and 10; they do not represent activity fractions.

Supplementary Table S2b. Complete raw-spectrum and five-window NNLS results for all source measurements.

Sample	Raw bg	Raw Am	Raw Cs	Raw Co	Raw residual (%)	Raw ranking	5W Am	5W Cs	5W Co	5W residual (%)	5W ranking
1	1.0000	1.0000	0.0000	0.0000	0.00	Am-241 > Co-60 > Cs-137	1.0000	0.0000	0.0000	0.00	Am-241 > Co-60 > Cs-137
2	0.9667	0.3782	0.0182	0.0201	0.94	Am-241 > Co-60 > Cs-137	0.3154	0.0000	0.0048	1.10	Am-241 > Co-60 > Cs-137
3	0.9986	0.1561	0.0000	0.0000	0.99	Am-241 > Co-60 > Cs-137	0.1490	0.0000	0.0000	5.68	Am-241 > Co-60 > Cs-137
4	0.9790	0.0941	0.0186	0.0000	1.42	Am-241 > Cs-137 > Co-60	0.0607	0.0000	0.0000	14.72	Am-241 > Co-60 > Cs-137
5	0.9855	0.0000	2.5394	0.0000	3.05	Cs-137 > Co-60 > Am-241	0.0000	2.5582	0.0000	4.32	Cs-137 > Co-60 > Am-241
6	1.0000	0.0000	1.0000	0.0000	0.00	Cs-137 > Co-60 > Am-241	0.0000	1.0000	0.0000	0.00	Cs-137 > Am-241 > Co-60
7	0.9868	0.0393	0.2329	0.0195	1.11	Cs-137 > Am-241 > Co-60	0.0228	0.2192	0.0121	2.63	Cs-137 > Am-241 > Co-60
8	0.9754	0.0578	0.1665	0.0047	0.97	Cs-137 > Am-241 > Co-60	0.0226	0.1401	0.0000	8.01	Cs-137 > Am-241 > Co-60
9	1.0181	0.0000	0.0000	2.6037	3.80	Co-60 > Cs-137 > Am-241	0.0000	0.0000	2.7476	4.27	Co-60 > Cs-137 > Am-241
10	1.0000	0.0000	0.0000	1.0000	0.00	Co-60 > Am-241 > Cs-137	0.0000	0.0000	1.0000	0.00	Co-60 > Cs-137 > Am-241
11	0.9827	0.0591	0.0108	0.1894	0.95	Co-60 > Am-241 > Cs-137	0.0246	0.0021	0.1802	1.45	Co-60 > Am-241 > Cs-137
12	0.9884	0.0483	0.0164	0.0791	0.99	Co-60 > Am-241 > Cs-137	0.0338	0.0034	0.0764	3.07	Co-60 > Am-241 > Cs-137
13	1.0255	0.0000	0.9353	0.7672	1.60	Cs-137 > Co-60 > Am-241	0.0270	0.9653	0.7794	1.12	Cs-137 > Co-60 > Am-241
14	0.9592	0.2175	0.0312	0.2183	0.88	Co-60 > Am-241 > Cs-137	0.1449	0.0025	0.2022	2.57	Co-60 > Am-241 > Cs-137
15	0.9869	0.0707	0.1907	0.1347	0.99	Cs-137 > Co-60 > Am-241	0.0568	0.1740	0.1297	1.02	Cs-137 > Co-60 > Am-241
16	1.0092	0.3412	0.6829	0.0029	0.97	Cs-137 > Am-241 > Co-60	0.3591	0.6922	0.0000	1.43	Cs-137 > Am-241 > Co-60
17	0.9623	0.2088	0.2631	0.2011	1.16	Cs-137 > Am-241 > Co-60	0.1538	0.2274	0.1877	2.41	Cs-137 > Co-60 > Am-241
18	0.9999	0.2550	0.0043	0.6387	1.11	Co-60 > Am-241 > Cs-137	0.2512	0.0013	0.6475	1.46	Co-60 > Am-241 > Cs-137
19	0.9658	0.1074	0.1894	0.2317	0.96	Co-60 > Cs-137 > Am-241	0.0475	0.1658	0.2147	1.14	Co-60 > Cs-137 > Am-241
20	1.0302	0.8680	0.6046	0.0043	1.15	Am-241 > Cs-137 > Co-60	0.9354	0.6221	0.0069	0.46	Am-241 > Cs-137 > Co-60
21	0.9840	0.0883	0.2432	0.3279	1.11	Co-60 > Cs-137 > Am-241	0.0560	0.2315	0.3265	1.62	Co-60 > Cs-137 > Am-241

Notes. Raw bg is the fitted coefficient of the measured background template. Raw-spectrum and five-window residuals are calculated in different representation spaces and should not be treated as identical statistical quantities.

Supplementary Table S3. Parametric Poisson resampling.

Sample	Expected	Am median	Am 95% interval	Cs median	Cs 95% interval	Co median	Co 95% interval	Ranking stability (%)	All positive (%)	R _{net} (%)	BCCS ε (%)
2	Am-241	0.3115	0.2784-0.3416	0.0000	0.0000-0.0116	0.0040	0.0000-0.0169	100.00	-	7.61	5.78
3	Am-241	0.1264	0.0845-0.1608	0.0000	0.0000-0.0041	0.0000	0.0000-0.0074	100.00	-	3.24	26.57
4	Am-241	0.0413	0.0000-0.0784	0.0000	0.0000-0.0032	0.0000	0.0000-0.0043	96.86	-	1.06	78.98
5	Cs-137	0.0000	0.0000-0.0000	2.5806	2.5611-2.6017	0.0000	0.0000-0.0000	100.00	-	132.28	1.35
7	Cs-137	0.0271	0.0000-0.0564	0.2196	0.2002-0.2367	0.0139	0.0000-0.0323	100.00	-	12.62	1.61
8	Cs-137	0.0550	0.0200-0.0911	0.1275	0.1109-0.1433	0.0000	0.0000-0.0043	99.84	-	7.54	6.61
9	Co-60	0.0000	0.0000-0.0000	0.0000	0.0000-0.0000	2.7635	2.7440-2.7824	100.00	-	134.58	1.57
11	Co-60	0.0231	0.0000-0.0504	0.0000	0.0000-0.0193	0.1809	0.1649-0.1943	100.00	-	9.66	1.35
12	Co-60	0.0226	0.0000-0.0516	0.0106	0.0000-0.0300	0.0743	0.0570-0.0903	99.84	-	4.79	3.84
13	Cs-137 + Co-60	0.0321	0.0000-0.0657	0.9485	0.9238-0.9739	0.7978	0.7746-0.8195	100.00	-	89.72	0.48
14	Am-241 + Co-60	0.1399	0.1117-0.1690	0.0129	0.0000-0.0320	0.1953	0.1769-0.2121	100.00	-	13.53	1.31
15	Cs-137 + Co-60	0.0608	0.0312-0.0900	0.1772	0.1566-0.1977	0.1282	0.1091-0.1468	100.00	-	16.85	1.20
16	Am-241 + Cs-137	0.3601	0.3277-0.3947	0.6874	0.6685-0.7038	0.0050	0.0000-0.0233	100.00	-	44.56	0.53
17	Am-241 + Cs-137 + Co-60	0.1719	0.1422-0.2020	0.2243	0.2031-0.2446	0.1881	0.1687-0.2077	-	100.00	24.68	1.31
18	Co-60 + Am-241	0.2379	0.2056-0.2687	0.0000	0.0000-0.0194	0.6531	0.6346-0.6683	100.00	-	38.23	0.78
19	Cs-137 + Co-60	0.0582	0.0286-0.0882	0.1637	0.1432-0.1842	0.2177	0.1984-0.2373	100.00	-	20.45	1.12
20	Am-241 + Cs-137	0.9468	0.9134-0.9790	0.6093	0.5878-0.6293	0.0193	0.0000-0.0380	100.00	-	54.82	0.87
21	Cs-137 + Co-60	0.0756	0.0448-0.1052	0.2131	0.1918-0.2347	0.3418	0.3218-0.3613	100.00	-	29.65	1.30

Notes. Each row summarizes 5,000 parametric Poisson resamples of the measurement and background spectra with the measured reference matrix held fixed. Ranking stability denotes correct top-rank recovery for single-source measurements and correct top-two recovery for binary mixtures. For ternary Sample 17, all-positive recovery is reported instead. The intervals include counting-statistical variation in the measurement and background spectra but do not include reference-acquisition, positioning, source-activity, gain, temperature, or environmental-background uncertainty.

Supplementary Table S4. Five-window reference responses.

Reference	W1 net cps	W2 net cps	W3 net cps	W4 net cps	W5 net cps	Reference condition
Am-241	1389.550	43.700	3.150	7.050	-8.650	Sample-1; 80 cm; 90 µCi
Cs-137	883.574	1364.590	656.126	239.686	26.960	Sample-6; 130 cm; 16 µCi
Co-60	408.499	731.173	548.195	1070.043	260.491	Sample-10; 120 cm; 9 µCi

Notes. Negative net window values arising from channel-wise background subtraction were retained and were not clipped to zero before NNLS analysis.

Supplementary Note 1. Validation using the opposing detector panel.

To examine whether the observed coefficient-ranking behavior was specific to the primary detector panel, an additional validation was performed using the opposing cRPM-54 detector panel (PN2). PN2 was operated at a photomultiplier high voltage of 1,425 V and an amplifier gain of 4.0, whereas the primary panel was operated at 1,475 V with the same gain. The additional data set comprised one background measurement, twelve single-source measurements, and nine mixed-source measurements. The nominal real time was 20 s, and the file-specific live times ranged from 18.052 to 20.000 s.

The lower-level discriminator setting differed slightly between the two panels. In the PN2 spectra, Channels 1 and 2 contained zero counts, whereas Channel 3 contained valid measured counts. Therefore, the PN2 analysis used Channels 3–256, rather than the Channels 4–256 range used for the primary panel. After file-specific live-time normalization, the PN2 background spectrum was subtracted channel by channel. The BCCS vector was then calculated at each threshold from Channel 3 through Channel 256, and NNLS decomposition was performed using panel-specific reference spectra.

Sample 4, acquired with Am-241 at 40 cm, Sample 8, acquired with Cs-137 at 110 cm, and Sample 11, acquired with Co-60 at 190 cm, were used as the fixed PN2 reference spectra. The remaining nine single-source spectra and nine mixed-source spectra were used for evaluation. The fitted coefficients are response-scaling factors relative to these panel-specific reference measurements and do not represent radionuclide activity fractions.

The correct radionuclide produced the largest coefficient in all nine non-reference single-source measurements. For all eight binary mixtures, the two actual constituents occupied the two largest coefficient ranks. All three coefficients were positive for the ternary mixture, Sample 17. The mixed-source BCCS residuals ranged from 0.54% to 4.22%, with a mean of 1.87%.

Parametric Poisson resampling was performed 5,000 times using the observed sample and background channel counts as independent Poisson means, while the measured PN2 reference matrix was held fixed. For the single-source and binary evaluation spectra, correct-ranking stability was 96.04% for Sample 1, 99.96% for Sample 2, 99.72% for Sample 5, and 99.10% for Sample 16; all other cases showed 100% stability. All three coefficients remained positive in 100% of the resamples for ternary Sample 17.

These results support the reproducibility of BCCS–NNLS for a detector panel operated under slightly different conditions when the panel-specific valid channel range, background spectrum, and reference spectra are used. This additional validation does not demonstrate calibration-free transfer of reference spectra between detector panels.

Supplementary Table S5. Complete BCCS–NNLS results for single-source measurements obtained with the opposing detector panel.

Sample	Measurement condition	NNLS coefficient	Coefficient ranking	R _{net} (%)	Residual (%)
1	Am-241, 420 cm	(0.0293 , 0.0008, 0.0036)	Am-241 > Co-60 > Cs-137	2.98	13.47
2	Am-241, 320 cm	(0.0419 , 0.0000, 0.0000)	Am-241 > (Cs-137 = Co-60)	4.13	14.39
3	Am-241, 220 cm	(0.0895 , 0.0000, 0.0000)	Am-241 > (Cs-137 = Co-60)	8.73	9.76
4	Am-241, 40 cm (reference)	(1.0000 , 0.0000, 0.0000)	Am-241 > (Cs-137 = Co-60)	95.45	0.00
5	Cs-137, 430 cm	(0.0154, 0.0821 , 0.0127)	Cs-137 > Am-241 > Co-60	6.68	4.36
6	Cs-137, 370 cm	(0.0110, 0.1231 , 0.0036)	Cs-137 > Am-241 > Co-60	8.74	3.71
7	Cs-137, 200 cm	(0.0109, 0.3522 , 0.0111)	Cs-137 > Co-60 > Am-241	23.67	1.11
8	Cs-137, 110 cm (reference)	(0.0000, 1.0000 , 0.0000)	Cs-137 > (Am-241 = Co-60)	64.54	0.00
9	Co-60, 440 cm	(0.0056, 0.0050, 0.2124)	Co-60 > Am-241 > Cs-137	4.60	4.30
10	Co-60, 380 cm	(0.0041, 0.0000, 0.2926)	Co-60 > Am-241 > Cs-137	5.79	3.12
11	Co-60, 190 cm (reference)	(0.0000, 0.0000, 1.0000)	Co-60 > (Am-241 = Cs-137)	19.22	0.00
12	Co-60, 30 cm	(0.0000, 0.0000, 13.9384)	Co-60 > (Am-241 = Cs-137)	259.44	2.77

Note. The spectra were processed using file-specific live-time normalization, channel-wise background subtraction, BCCS over threshold Channels 3–256, and NNLS. R_{net} was calculated over Channels 3–256. Samples 4, 8, and 11 were used as the panel-specific reference spectra.

Supplementary Table S6. Complete BCCS–NNLS results for mixed-source measurements obtained with the opposing detector panel.

Sample	Measurement condition	NNLS coefficient	Coefficient ranking	R _{net} (%)	Residual (%)
13	Cs-137 359 cm + Co-60 359 cm	(0.0227, 0.1222 , 0.3139)	Co-60 > Cs-137 > Am-241	15.61	3.01
14	Am-241 208 cm + Co-60 208 cm	(0.1057 , 0.0056, 0.8385)	Co-60 > Am-241 > Cs-137	26.33	1.45
15	Cs-137 134 cm + Co-60 134 cm	(0.0132, 0.6967 , 1.9059)	Co-60 > Cs-137 > Am-241	82.59	0.90
16	Am-241 332 cm + Cs-137 332 cm	(0.0523 , 0.1346 , 0.0077)	Cs-137 > Am-241 > Co-60	13.64	2.84
17	Am-241 + Cs-137 + Co-60, each at 186 cm	(0.1388 , 0.3952 , 0.9962)	Co-60 > Cs-137 > Am-241	57.49	0.82
18	Am-241 285 cm + Co-60 342 cm	(0.0609 , 0.0000, 0.3455)	Co-60 > Am-241 > Cs-137	12.31	2.17
19	Cs-137 128 cm + Co-60 210 cm	(0.0059, 0.7788 , 0.8328)	Co-60 > Cs-137 > Am-241	66.57	0.54
20	Am-241 410 cm + Cs-137 322 cm	(0.0392 , 0.1473 , 0.0000)	Cs-137 > Am-241 > Co-60	12.91	4.22
21	Cs-137 272 cm + Co-60 179 cm	(0.0133, 0.4344 , 0.4893)	Co-60 > Cs-137 > Am-241	38.31	0.89

Note. The two actual constituents occupied the two largest coefficient ranks for all eight binary mixtures. Sample 17 was evaluated as a ternary decomposition-consistency case, for which all three fitted coefficients were positive. The mean residual for the nine mixed-source measurements was 1.87%.

Supplementary Table S7. Poisson-resampling results for the opposing detector panel.

Sample	Expected	Am median (95% interval)	Cs median (95% interval)	Co median (95% interval)	Ranking stability (%)	All positive (%)
1	Am-241	0.0280 (0.0201–0.0353)	0.0000 (0.0000–0.0080)	0.0002 (0.0000–0.0301)	96.04	–
2	Am-241	0.0408 (0.0315–0.0484)	0.0000 (0.0000–0.0044)	0.0000 (0.0000–0.0231)	99.96	–
3	Am-241	0.0882 (0.0790–0.0968)	0.0000 (0.0000–0.0000)	0.0000 (0.0000–0.0229)	100.00	–
5	Cs-137	0.0158 (0.0088–0.0226)	0.0810 (0.0697–0.0914)	0.0127 (0.0000–0.0505)	99.72	–
6	Cs-137	0.0118 (0.0045–0.0188)	0.1206 (0.1099–0.1308)	0.0030 (0.0000–0.0402)	100.00	–
7	Cs-137	0.0112 (0.0041–0.0184)	0.3511 (0.3393–0.3617)	0.0109 (0.0000–0.0485)	100.00	–
9	Co-60	0.0052 (0.0000–0.0119)	0.0049 (0.0000–0.0168)	0.2117 (0.1760–0.2444)	100.00	–
10	Co-60	0.0032 (0.0000–0.0100)	0.0000 (0.0000–0.0112)	0.2896 (0.2564–0.3194)	100.00	–
12	Co-60	0.0000 (0.0000–0.0000)	0.0000 (0.0000–0.0000)	13.9390 (13.8784–13.9987)	100.00	–
13	Cs-137 + Co-60	0.0226 (0.0158–0.0295)	0.1220 (0.1096–0.1347)	0.3141 (0.2750–0.3527)	100.00	–
14	Am-241 + Co-60	0.1055 (0.0983–0.1127)	0.0055 (0.0000–0.0184)	0.8385 (0.7977–0.8733)	100.00	–
15	Cs-137 + Co-60	0.0131 (0.0053–0.0209)	0.6967 (0.6816–0.7126)	1.9062 (1.8604–1.9522)	100.00	–
16	Am-241 + Cs-137	0.0527 (0.0454–0.0599)	0.1328 (0.1219–0.1425)	0.0081 (0.0000–0.0449)	99.10	–
17	Am-241 + Cs-137 + Co-60	0.1387 (0.1311–0.1463)	0.3956 (0.3816–0.4092)	0.9954 (0.9531–1.0379)	–	100
18	Am-241 + Co-60	0.0608 (0.0528–0.0682)	0.0000 (0.0000–0.0070)	0.3446 (0.3141–0.3722)	100.00	–
19	Cs-137 + Co-60	0.0060 (0.0000–0.0137)	0.7786 (0.7643–0.7927)	0.8332 (0.7909–0.8747)	100.00	–
20	Am-241 + Cs-137	0.0394 (0.0315–0.0470)	0.1470 (0.1354–0.1576)	0.0000 (0.0000–0.0099)	100.00	–
21	Cs-137 + Co-60	0.0133 (0.0061–0.0202)	0.4344 (0.4212–0.4478)	0.4892 (0.4494–0.5295)	100.00	–

Note. Each row summarizes 5,000 parametric Poisson resamples of the measurement and background spectra, with the measured PN2 reference matrix held fixed. Ranking stability denotes correct top-rank recovery for single-source measurements and correct top-two recovery for binary mixtures. For ternary Sample 17, the percentage of resamples yielding positive coefficients for all three radionuclides is reported. Reference Samples 4, 8, and 11 were excluded because they defined the fixed reference matrix. The intervals represent the 2.5th–97.5th percentiles.